

Saurav Raj Kunwar

saunkunwa@ttu.edu | (929) 332-1743 | Lubbock, TX 79409 | Nationality: Nepali

Profile

Applied economist working at the intersection of climate risk, agricultural productivity, and decision-making under uncertainty. Uses quasi-experimental identification, causal inference, and geospatial climate data to quantify how uninsured weather risk drives resource misallocation and erodes resilience, including work on developing-country household panels for Malawi and Ethiopia. Peer-reviewed publications in leading field outlets, including the *American Journal of Agricultural Economics* and the *Annual Review of Resource Economics*. Fluent in English, Nepali, and Hindi.

Education

University of Illinois Urbana–Champaign May 2026

PhD, Agricultural and Applied Economics

Dissertation: *Three Essays on Environmental Services Valuation and Agricultural Productivity*

Committee: Teresa Serra (chair), Robert Chambers, Scott Irwin, Laura Gentry, Gary Schnitkey

University of Georgia May 2020

MS, Agricultural and Applied Economics

Thesis: *Understanding Profitability of Georgia Blueberry Growers Adopting a Stochastic Approach*

Tribhuvan University, Nepal Aug. 2017

BS, Agriculture (Nepal Government Scholarship, 2013–2017)

Professional Experience

Postdoctoral Researcher June 2026–Present

Texas Tech University, Department of Agricultural and Applied Economics

Lubbock, TX

- Research on agricultural productivity, Total Resource Productivity, and producer decision-making under uncertainty, working with Keith Fuglie (World Bank senior consultant) and Krishna Paudel.
- Co-PI on a \$1,000,000 USDA-NRCS Conservation Innovation Grant (On-Farm Trials) evaluating the economic value of irrigation information for Texas High Plains cotton; applies an expected-value-of-information framework tested through a randomized on-farm experiment with producer cooperators. Submitted July 2026.

Research Scholar Jan.–May 2023

University of Maryland, Department of Agricultural and Resource Economics

College Park, MD

- Frontier estimation of production and environmental efficiency under uncertainty, with Robert Chambers.

Graduate Research Assistant 2020–2026

University of Illinois Urbana–Champaign

Urbana, IL

- Field-level analysis of conservation adoption, carbon services, and yield resilience to weather shocks using Precision Conservation Management farm-level data.

Climate-Relevant Research

Weather Risk, Resource Misallocation, and Smallholder Productivity: Evidence from Malawi and Ethiopia

Solo-authored, in progress

- Combines LSMS-ISA household panel data for Malawi with satellite climate data (CHIRPS, ERA5-Land); uses the 2015/16 El Niño drought as a quasi-experiment.
- Methods: ACF production-function estimation, Olley–Pakes decomposition, event-study identification.
- Preliminary finding: TFP dispersion rose approximately 30% following the drought and persisted for three years, indicating that uninsured weather risk drives resource misallocation among smallholders.

Conservation Practices and U.S. Corn Yield Resilience to Weather Shocks

With R. G. Chambers, L. Gentry, and T. Serra — working paper

- Panel stochastic frontier analysis of how conservation practices affect yield resilience to weather shocks.

What Are the Carbon Services from Cover Crop Adoption Worth from Farmers' Perspective?

Kunwar, Chambers, Gentry & Serra (2026), Am. J. Agric. Econ. 108(3): 901–925

- Shadow pricing of carbon sequestration and GHG emissions from cover crop adoption, with soil organic matter treated as dynamic capital.

Additional Peer-Reviewed Publications

- Chambers, R. G., **Kunwar, S. R.**, Quiggin, J., Serra, T., & Wang, S. (2025). Estimating Production and Environmental Efficiency under Uncertainty. *Annual Review of Resource Economics*, 17: 85–104.
- **Kunwar, S. R.**, Fonsah, E. G., & Escalante, C. L. (2025). An Assessment of Profitability Using Monte Carlo Simulation Approach: A Case of Georgia Blueberry Industry. *Journal of Food Distribution Research*.
- Fonsah, E. G., & **Kunwar, S. R.** (2024). Economics of Satsuma Citrus Production Using Drip Irrigation and Frost Protection in the United States. *Agricultural Research & Technology*, 28(4): 1–11.
- Fonsah, E. G., **Kunwar, S. R.**, & Diaz-Perez, J. C. (2023). Economic Evaluation of Bell Pepper Production Under Colored Shade Nets System in the Southeast Region of the USA. *African Journal of Agricultural Economics and Rural Development*, 11(1): 1–7.
- Bogati, S., **Kunwar, S. R.**, & Amgain, L. P. (2021). Assessment of Nutrient Expert Hybrid Maize Model on Productivity and Profitability of Winter Maize in Terai Region of Eastern Nepal. *Heliyon*, 7(10): e08096.
- **Kunwar, S. R.**, Fonsah, E. G., & Ramirez, O. (2021). Understanding Profitability of Georgia Blueberry Growers Adopting a Stochastic Approach. *Journal of Food Distribution Research*, 52: 39–45.
- **Kunwar, S. R.**, Bogati, S., Fonsah, E. G., & Amgain, L. P. (2019). Economic Assessment of Adopting Nutrient Expert Wheat Model vs. Conventional Wheat Fertilizer Application Management in Morang, Nepal. *Journal of Agricultural Studies*, 7(3): 38–48.

Research in progress: Comprehensive Resource-Extended State Agricultural TFP (with K. Paudel, K. Fuglie); Agricultural TFP Convergence in the U.S. and the Innovation Channel of Public R&D and Patents (with G. Seo, K. Paudel, K. Nelson, K. Fuglie).

Grant Activity

- Co-PI, *Economic Value of Irrigation Information for Texas High Plains Cotton*, USDA-NRCS Conservation Innovation Grant (On-Farm Trials), \$1,000,000. Submitted July 2026.

Technical Skills

Programming / modeling: R, Stata, Python, GAMS, SAS, SPSS; Git/GitHub, LaTeX; reproducible, version-controlled analytical workflows.

Quantitative methods: panel and spatial econometrics; causal inference (difference-in-differences, event study, instrumental variables); system GMM; control-function estimators; structural modeling; production-function and stochastic-frontier estimation; Monte Carlo simulation; cost–benefit analysis.

Data: developing-country household panel surveys (LSMS-ISA); geospatial and remote-sensing products (CHIRPS, ERA5-Land); watershed- and county-scale administrative data.

Languages: English (fluent), Nepali (native), Hindi (fluent).